

Pascal Jutras-Dubé

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Education

PhD Computer Science	Purdue University	2023 – expected May 2028
MSc Computer Science	Université de Montréal	2020 – 2022
BSc Mathematics and Computer Science	Université de Montréal	2017 – 2020

Experience

- Research Assistant**, Purdue University Jan 2023 – Present
- **Sticky Jump Diffusions** [arXiv] [code]: Introduced sticky jump diffusions unifying masked, continuous, and hybrid diffusion. Time-marginals mix a continuous density with atoms on a discrete set, so graded information stays available to a provably discrete sampler. I derived the exact time reversal by flux balance, and proved its intractable reverse jump intensity equals a conditional expectation of forward-known quantities, estimated by a Rao–Blackwellized reweighting of one cross-entropy classifier with no auxiliary network. Training is simulation-free via inverse-CDF sampling of jump times and closed-form Gaussian convolutions. Cuts CIFAR-10 FID 15.9 → 14.6 and lifts Sudoku exact-match accuracy 47% → 96% at 4× faster convergence over the strongest prior hybrid under matched compute.
 - **One-step diffusion samplers** [arXiv] [code]: Learned a step-conditioned ODE so one large step reproduces a many-step trajectory, and derived a deterministic-flow importance weight that keeps evidence (ELBO) estimates stable where standard integrators break down. Matches multi-step sample quality and estimator accuracy at 1/256 of the network evaluations.
- Teaching Assistant**, Purdue University Jan 2023 – Present
- TA for graduate Statistical Machine Learning, incl. guest lectures on diffusion models.
- Research Intern**, National Bank of Canada Jun 2021 – Nov 2022
- Built a quantitative scoring framework (record linkability, information gain) measuring the privacy–utility frontier of customer data-release strategies; published at HICSS 2024 and in the *Journal of Business Analytics* (2025).
- Software Engineer Intern**, GRAITEC May 2019 – Aug 2019
- Shipped full-stack features for a production woodworking CAD application in an agile team.

Selected Publications

- Jutras-Dubé, P., Pynadath, P., Lu, J., Gao, Y., & Zhang, R.
Sticky Jump Diffusions: A Unifying View of Masked, Continuous, and Hybrid Diffusion. NeurIPS, 2026. [code]
- Jutras-Dubé, P., Zhang, J., Wang, Z., & Zhang, R.
One-Step Diffusion Samplers via Self-Distillation and Deterministic Flow. AISTATS, 2026. [code]
- Lee, S., Jutras-Dubé, P., Wen, B., & Zhang, R.
Neural Discrete Controlled Monte Carlo Samplers. ProbML Workshop @ ICML, 2026
- Jutras-Dubé, P., Al-Khasawneh, M. B., Yang, Z., Bas, J., Bastin, F., & Cirillo, C.
Copula-Based Transferable Models for Synthetic Population Generation. Transportation Research Part C, 2024. [code]
- Jutras-Dubé, P., Zhang, R., & Bera, A.
Adaptive Planning with Generative Models under Uncertainty. IROS, 2024. [code]
- Full list: [Google Scholar](#).

Skills & Interests

Interests: Generative modeling & sampling, diffusion language models, transformers & sequence models, SDEs and jump processes, Monte Carlo, Bayesian inference, optimization

Engineering: Python, JAX, PyTorch, NumPy/pandas, Hydra, Weights & Biases, Slurm, Git

Additional Information

Awards: Fin-ML NSERC CREATE Scholarship national training program in machine learning for quantitative finance (2020); UdeM Graduate Scholarship (2021); DIRO Excellence (4×, 2020–2022)

Reviewer: NeurIPS 2026, ICML 2026 (Silver Reviewer Award), ICLR 2025–2026, AISTATS 2026, AAAI 2027

Languages: French (native), English (fluent)